35 Vapour deposited coatings and thermal spraying

Vapour deposition processes are of two kinds: physical vapour deposition (PVD) and chemical vapour deposition (CVD). Process details and references for elements are given in Tables 35.1 and 35.2, for oxides in Tables 35.3 and 35.4, for nitrides in Tables 35.5 and 35.6, and for carbides in Tables 35.7 and 35.8. Section 35.3 discusses thermal spraying (*see* Chapter 33 for relevant standards).

35.1 Physical vapour deposition

Physical vapour deposition processes use a physical effect such as evaporation or sputtering to transport material, usually a metal, from a source to the substrate to be coated. If the material is transformed (e.g. into a carbide) during transport, then the process is described as reactive. All PVD processes are carried out in a relatively high vacuum (i.e. pressure $<10^{-4}$ Torr).

35.1.1 Evaporation (E)

The substrate to be coated is placed in a vacuum chamber with a line-of-sight to the source which is a pool of molten material. The pool is heated either by an electron beam or an arc or by resistance heating. The electron beam method is best for source materials with a high melting point.

REACTIVE EVAPORATION (RE)

The same process as evaporation except that the zone through which the evaporated species is passing contains a very small concentration of a reactive gas, usually a hydrocarbon, N_2, NH_3 or O_2. This results in the deposition of the carbide, nitride or oxide of the source material.

ACTIVATED REACTIVE EVAPORATION (ARE)

The same process as reactive evaporation except that an electrically excited plasma is established in the region where the evaporated species encounters the reactive gas.

ION PLATING (IP)

The same process as evaporation except that substrate is biassed negatively with respect to the source. This usually results in a plasma region around the substrate which can be enhanced by r.f. coupling. The ionised species actually represents only a small fraction of the total material flux.

REACTIVE ION PLATING (RIP)

The same process as ion plating except that a reactive gas (usually a hydrocarbon, N_2, NH_3 or O_2) is introduced into the flux travelling towards the substrate so that the carbide, oxide or nitride of the source material is formed.

35.1.2 Sputtering (S)

A process in which material is transferred from a target and deposited on a substrate by means of ionic bombardment of the target. Usually the ion bombardment is achieved by establishing a d.c. or r.f. plasma at the surface of the target, although ion guns can also be used.

SPUTTER ION PLATING (SIP)

The same process as sputter plating except that the substrate has an electrical negative bias, d.c. and r.f. possible (for non conductive materials), so that it is bombarded with positive ions as well as the neutral flux created by the sputtering.

REACTIVE SPUTTER PLATING (RSP)

The same process as sputter plating except that a reactive gas is introduced into the plasma region so that the transported target material reacts before it arrives at the substrate. For example, the conversion of Ti into a deposit of TiN by the introduction of nitrogen.

MAGNETRON SPUTTERING (MS)

The same process as sputter plating except that a magnetic field is used to direct the ion flux at the surface of the target.

35.1.3 Ion cleaning

An argon plasma may be used to bombard a surface with excited argon atoms. These transfer their momentum to atoms in the surface and some of these are ejected from the surface. This has the effect of gradually removing material from a surface, hence cleaning the surface. Ion cleaning is often used as a pretreatment in PVD coating.

35.2 Chemical vapour deposition

Chemical vapour deposition processes use the vapour phase to transport reactive material to the surface of a substrate where a chemical reaction occurs to form a coating, e.g. tantalum by the reaction: $TaCl_5 + H_2 \rightarrow Ta + 5HCl$. Normally the substrate is heated to activate the reaction. In many CVD processes the substrate takes no part in the formation of the coating, but in *diffusion coatings* the substrate takes part in the formation of the coating (e.g. aluminising).

CVD processes can be divided into two groups, hot wall and cold wall, as follows:

A hot-wall reactor is a chamber which is heated from the outside so that both the chamber and the work pieces reach the reaction temperature and so become coated. Consequently there is a certain amount of redundant plating which is undesirable with expensive materials.

A cold-wall reactor is a chamber in which only the work pieces are heated, usually by induction. Only the work pieces become coated.

CVD processes are a rather diffuse group because the physical conditions for each process differ widely so that there are no standard items of equipment as there are for PVD. Many CVD processes use a reduced pressure to improve throwing power (cf. PVD).

In the references *CVD* stands for the International Chemical Vapor Depositions Conference Proceedings published by The Electrochemical Society, Vapour Deposition in the refs. refers to that title by Powell, Oxley and Blocher, J. Wiley, N.Y. 1965. Process details Tables 35.2, 35.4, 35.6, 35.8.

Plasma Activated or Enhanced Chemical Vapour Deposition (PACVD or PECVD)

The same process as CVD except that a d.c. or r.f. plasma is established around the substrate, usually to permit a low coating temperature.

Metal Organic Chemical Vapour Deposition (MOCVD)

The same process as CVD except that organometallic chemicals (e.g. trimethylgallium) are used as the feed reagents. Many organometallics have a high vapour pressure and a low decomposition temperature so that coatings can be formed at low substrate temperatures. It is particularly used for the growth of epitaxial layers.

Chemical Vapour Infiltration (CVI)

The same process as CVD except that the substrate is a porous and usually fibrous material into which the deposited material is infiltrated. It is an extremely slow process, but it leads to composites of low final porosity.

Table 35.1 PHYSICAL VAPOUR DEPOSITION OF ELEMENTS AND METALLIC ALLOYS

Element	*Process**	*Coating thickness* (μm)	*Typical applications*	*Ref.*
Ag	S	0.02–0.35	Window glass	2
	RIP		Photographic materials	6
			Foil bearings	31
Al**	E	~1	IC metallisation	1
	S			3
	MS	~1	Mirror coating	8
Au**	IP	0.2		4
	E	0.02–0.1	Window glass, electron microscopy	7
	S	1.4	Coating micro-spheres for laser fusion	5
Bi	S		Undercoat for photographic materials	6
	E	~1		
C	TS	0.05	Conductive layers for electron	32
	Arc	350	microscopy samples	19
(Co (Cr/Al))***	E		Turbine blades	14, 15
Cr	3	0.1–0.6	Ceramics	11
			Plastics	12
			Ni/Cr for window glass	16
			Switch contacts	
Cu	IP	0.2	Window glass coating	4
	S	0.02	Metallisation of plastics prior	11
	MS	0.01–1.0	to electroplating	13
Mo	IBS		Multi-level metallisation for LSI	17
	S	0.15		18
Nb	S	0.4	Superconductors	22
Ni	E, S	—	—	25
Pb	E (EG)	0.4	Pb/oxide/Pb Josephson junctions	21
Pd	E	0.06	Thermoelectric power	21
	E (EG)	0.1–0.15	Contact layer	23, 24
Pt	MS	1.0	Schottky barrier solar cells	10
	E			—
Si	IBS	0.4	Integrated circuits	—
	E			
Ta	S, E	6–20	Corrosion resistant layer	26, 27
Ti	S	—	—	12
Ti–W	MS	500–5 000 Å	Resistance films	—
V	E	0.086	—	30
W	S	500–5 000 Å	Barrier layer for ICs	28, 29

* ARE — Activated reactive evaporation.
** also used for compact disc metallising.
*** Typically MCrAly *alloys* M = Co, Ni.
E — Evaporaiton.
AE — Arc evaporation.
EG — Electron gun.
IBS — Ion beam sputtering.
IP — Ion Plating.
MS — Magnetron sputtering.
RIS — Reactive ion sputtering.
RIP — Reactive ion plating.
RE — Reactive evaporation.
RAE — Reactive arc evaporation.
RSP — Reactive sputtering.
S — Sputtering.
SIP — Sputter ion plating.
TS — Triode sputtering.

REFERENCES TO TABLE 35.1

1. R. J. Hill, 'Physical Vap. Dep.,' Airco Temescal, Berkeley, California, 1976, p. 114.
2. T. Abe and T. Yamashina, *Thin Solid Films*, 1975, **30**, 19.
3. C. R. Fuller and P. B. Ghate, *Thin Solid Films*, 1979, **64**, 25.
4. T. Spalvins, *Thin Solid Films*, 1979, **64**, 143.
5. L. Buene *et al.*, *Thin Solid Films*, 1980, **65**, 247.
6. S. K. Sharma and J. Spitz, *Thin Solid Films*, 1980, **65**, 339.

7. A. T. Lowe and C. D. Hosford, *J. Vac. Sci. Tech.*, 1979, **16**, 197.
8. A. J. Learn, *J. Electrochem. Soc.*, 1976, **123**, 894.
9. A. Aronson and S. Weinig, *Vacuum*, 1977, **27**, 151.
10. S. Schiller *et al.*, *J. Vac. Sci. Tech.*, 1977, **14**, 813; 1975, **12**, 858.
11. S. Hurwitt, Trans. Conf. Sputter, Mats. Res. Corp., Orangeberg, NY, 1974, p. 31.
12. D. W. Hoffmann and J. A. Thornton, *Thin Solid Films*, 1977, **40**, 355.
13. T. Tsutada and N. Hosokawa, *J. Vac. Sci. Tech.*, 1979, **16**, 348.
14. D. H. Boone *et al.*, *Thin Solid Films*, 1979, **64**, 299.
15. C. J. Spengler and S. Y. Lee, *Thin Solid Films*, 1979, **64**, 263.
16. Matsushita Electric Works, JP 5 476 972.
17. P. H. Schmidt *et al.*, *J. Appl. Phys.*, 1973, **44**, 1833.
18. J. Nagano, *Thin Solid Films*, 1980, **67**, 1.
19. Y. Sakai *et al.*, *J. Nucl. Mater.*, 1988, 155.
20. M. Murahami, *Thin Solid Films*, 1980, **69**, 253.
21. G. Wedler and R. Chander, *Thin Solid Films*, 1980, **65**, 53.
22. C. T. Wu, *Thin Solid Films*, 1979, **64**, 103.
23. D. J. Sharp, *J. Vac. Sci. Tech.*, 1979, **16**, 204.
24. V. Köster *et al.*, *Thin Solid Films*, 1980, **67**, 35.
25. S. Schiller *et al.*, *J. Vac. Sci. Tech.*, 1975, **12**, 858.
26. S. Kashu *et al.*, *J. Vac. Sci. Tech.*, 1972, **9**, 1399.
27. J. Spitzel and J. Chevallier, *CVD*, 1975, **V**, 204.
28. Kossowsky, *Electrochem. Soc. Ext. Abst.*, 1977, **77-2**, 429.
29. J. B. Bindell and T. C. Tisone, *Thin Solid Films*, 1974, **23**, 31.
30. A. Borodziuk-kulpa *et al.*, *Thin Solid Films*, 1980, **67**, 21.
31. C. Dellacorte *et al.*, Nasa Lewis Research Center, Nasa Tech. Memo. NASA TM-100783, 1988.
32. A. G. Fitzgerald *et al.*, *Carbon*, 1988, **26** (2), 229.

Table 35.2 CHEMICAL VAPOUR DEPOSITION OF ELEMENTS

Elements	*Process*	*Temp* (°C)	*Thickness* (μm)	*Typical applications*	*Ref.*
Al	Al R or Al HR_2 thermal decomp. (R usually isobutyl)	200–600	50	Impervious layer on plastics	6, 9
	$Al + AlX^3$ (X = halide)	700–1 000	20–100	Aluminising steel and	10
	$AlCl_3 + H_2$	900–1 000		super alloys	15
As	AsH_3 thermal decomp.	230–300	—	GaAlAs lasers	7
	AsH_3 plasma assisted				1
B	B_2H_6 thermal decomp.	400–700	—	B fibres	13
	BBr_3 thermal decomp.	1 000–1 300			
	$BBr_3 + H_2$	600–1 100		High purity boron	5
	$BCl_3 + H_2$ (also with laser excitation)	1 000–1 500	5–1 000	Jewelled bearings, fibres	2, 5, 12, 72
C (graphite)	CH_4, C_2H_6 thermal decomp.	2 000–2 500	500–3 000	Free-standing components	14
C (carbon)	CH_4, C_2H_6 thermal decomp.	900–1 200	1–50	Barrier layers, C–C composites	3, 4, 11
C (diamond and diamond-like carbon)	CH_4, C_3H_8 microwave plasma assisted	200–800	0.1–20	Anit-reflectance coating	70, 20
Co	$Co(acac)_2$* thermal decomp.	300–500	1.5	—	17
Cr	CrI_2 thermal decomp.	900–1 400	—	High purity chromium	19
	Cr dicumene	300–400	25–50	Coating nuclear fuel elements	14
	$Cr + NH_4Cl$ pack	950–1 050	20–50	Coating turbine blades	16
Cu	$Cu(acac)_2$* thermal decomp.	260–450	—	—	18
Ge	$GeCl_4 + H_2$	600	—	—	22
	GeH_4 thermal decomp.	400–900		Epitaxial semiconductor layer	21
Hf	HfX_4 thermal decomp. (X = I or Br)	—	—	—	25

(*continued*)

Table 35.2 CHEMICAL VAPOUR DEPOSITION OF ELEMENTS—*continued*

Elements	*Process*	*Temp* (°C)	*Thickness* (μm)	*Typical applications*	*Ref.*
Ir	$Ir(CO)_2Cl_2$ thermal decomp.	600	—	—	28
Mo	$MoX_5 + H_2$ (X = Cl, F)	650–1 400	8	Thin film resistors	26, 30
	$(Mo(CO)_6 + H_2$	300–600	0.2	Coating nuclear fuels	29
	$Mo(CO)_6$ plasma			Anti-reflection on surfaces for photochem. conversion	27, 24, 23
Nb	$NbCl_5 + H_2$	1 000–1 400	5–50	Nuclear fuel cladding	32
Ni	$Ni(CO)_4$ thermal decomp.	180–200	1–1 000	Vapour forming	33
	$Ni(acac)_2$* thermal decomp.	250–450		Coating of uranium	34, 38
	$Ni(CO)_4$ plasma assisted				24
	$Ni(CO)_4$ laser assisted		0.055	Localised deposit	31
Pb	$PbEt_4$ thermal decomp.	300–500	—	—	35
Pd	$Pd(acac)_2$* thermal decomp.	350–450	—	—	39
Pt	$Pt(CO)_2Cl_2$ thermal decomp.	600	—	—	46
	$Pt(acac)_2$* thermal decomp.	350–450			40
Re	$ReF_5 + H_2$	850–1 100	—	—	—
	$ReCl_5 + H_2$	600–1 200			42
	$ReOCl_4$ thermal decomp.	1 250–1 500			37
	$ReCl_5$ thermal decomp.	1 100–1 300		Thermionic emitters	41
Rh	$Rh(tfa)_3$† $+ H_2$	250	—	—	38
	$Rh(CO)_2Cl_2$ thermal decomp.	600			46
Ru	$Ru(CO)_5$ thermal decomp.	200	—	—	47
	$Ru(CO)_2Cl_2$ thermal decomp.	600			46
Sb	$SbCl_3$ thermal decomp.	500–600	—	—	49
	SbH_3 thermal decomp.	~150		Prod. of high purity Sb	48
Si	SiH_4 thermal decomp.	300–1 200	—	—	51, 71
	$SiCl_4 + H_2$	900–1 800		Prod. of polycryst. Si	
	$SiHCl_3 + H_2$	950–1 250	30	Solar cells	50
	SiH_4 plasma decomp.	300–1 150		Semiconductor layers	44
	Siliconising Ni alloys			Corrosion-resistant layers on gas turbine parts	45, 43
Sn	$Sn + NH_4Cl/MgO$ pack diffusion	450–750	10	Corrosion and wear resistant coating	53
Ta	$TaCl_5 + H_2$	900–1 100	5–20	Corrosion-resistant layers	54
				Thin film resistors	58
				Impregnation of C fibres	59
Ti	$TiBr_4$ thermal decomp.	1 100–1 400	—	Metal production	55
	TiI_4 thermal decomp.	1 200–1 500			56
U	UI_6 thermal decomp.	~1 500	—	—	57
V	$VCl_4 + H_2$	800–1 000	—	—	60
	VI_2	1 000–1 200	10–50		69, 61
W	$WF_6 + H_2$	400–700	1–1 000	Thin wall components	62
	$WCl_6 + H_2$	600–700	1–1 000	Erosion-resistant	66, 63,
	$W(CO)_6$ thermal decomp.	350–600		coatings on rocket motors	64
				Electron emission surfaces	67, 68
Zr	ZrI_4 thermal decomp.	1 000–1 500	5–50	Nuclear fuel cladding	65

* acac = acetylacetonate. † tfa = trifluoro-acetylacetonate.

REFERENCES TO TABLE 35.2

1. J. C. Knights and J. E. Matan, *Solid State Corrosion*, 1977, **21**, 983.
2. R. M. Mehalso and R. J. Diefendorf, *CVD*, 1975, **V**, 84.
3. R. P. Gower and J. Hill, *CVD*, 1975, **V**, 114.
4. W. F. Knippenberg *et al., Philips Tech. Rev.*, 1977, **37**, 189.
5. D. R. Stern and L. Lynds, *J. Electrochem Soc.*, 1958, **105**, 676.
6. H. O. Pierson, *Thin Solid Films*, 1977, **45**, 257.
7. K. Tamaru, *J. Phys. Chem.*, 1955, **59**, 777.
8. 'Vapor Deposition', p. 283.
9. E. R. Breining *et al.*, Ger. 1 235 106, 1967.
10. 'Vapor Deposition', p. 277.
11. J. Chin *et al., CVD*, 1975, **VI**, 364.
12. H. E. Hintermann *et al., Ext. Abs. Electrochem. Soc.*, 1973, **73/2**, 218.
13. R. B. Reeves and J. J. Gelhardt, *SAMPE*, 1979, **10** (1), 13.
14. J. H. Oxley *et al., Ind. and Eng. Chem.: Prod. Res. and Dev.*, 1962, **1**, 102.
15. L. H. Marshall, US 1 893 782, 1933.
16. H. M. J. Mazille, *Thin Solid Films*, 1980, **65**, 67.
17. E. J. Jablonwski, *Cobalt*, 1962, **14**, 28.
18. P. Palovlyk, US 2 704 728, 1955.
19. Powell *et al.*, 'Vapor Deposition', p. 290.
20. A. Sawabe *et al., Appl. Phys. Lett.*, 1987, **50**, 728.
21. D. J. Dumin *et al., RCA Rev.*, 1970, **31**, 620.
22. E. C. Cave and B. B. Czorny, *RCA Rev.*, 1963, **24**, 523.
23. B. O. Seraphin, *J. Vac. Sci. Tech.*, 1979, **16**, 193.
24. H. F. Sterling *et al., Vide*, 1966, **21**, 80.
25. F. B. Litton, *J. Electrochem. Soc.*, 1951, **98**, 488.
26. J. E. Cline and J. Wulff, *J. Electrochem. Soc.*, 1951, **98**, 385.
27. J. J. Lander and L. H. Germer, Am. Inst. Min. Met. Eng.: Inst. Met. Div. Met. Tech., **14**, 6. Tech. Prod. 2259, 1947.
28. J. A. M. van Liempt, *Metallwerkschaft*, 1932, **11**, 357.
29. K. Hieber and M. Stolz, *CVD*, 1975, **V**, 436.
30. R. R. Jaeger and S. T. Cohen, *CVD*, 1972, **III**, 500.
31. S. D. Allen and M. Bass, *J. Vac. Sci. Tech.*, 1979, **16**, 431.
32. W. A. Jenkins and H. W. Jacobson, US 3 020 148, 1962.
33. L. W. Owen, *J. Less-Common Met.*, 1962, **4**, 35.
34. E. C. Marboe, US 2 430 520, 1947.
35. R. N. Meinert, *J. Am. Chem. Soc.*, 1933, **55**, 979.
36. E. H. Reerin, *Z. f. Anorg. Chem.*, 1928, **173**, 45.
37. A. N. Zelikman *et al., Russian Metall.*, 1963, **4**, 120.
38. R. L. Van Hemert *et al., J. Electrochem. Soc.*, 1965, **112**, 1123.
39. C. F. Powell, J. H. Oxley and J. M. Blocher, in 'Vapor Deposition', Wiley, NY, 1968.
40. 'Vapor Deposition', p. 314.
41. L. Yang *et al., CVD*, 1972, **III**, 253.
42. P. J. Sherwood, *CVD*, 191073, **III**, 728.
43. A. R. Nicoll *et al., Thin Solid Films*, 1974, **64**, 321.
44. H. F. Sterling and R. C. G. Swan, *Solid State Electronics*, 1965, **8**, 653.
45. W. G. Townsend and M. E. Uddin, *Solid State Electronics*, 1973, **16**, 39.
46. E. H. Reerin, US 1 818 909, 1931.
47. Smelin, 'Handbuch der Anor. Chem.', 8th edn, Verlag Chemie Berlin, p. 63, 1938.
48. K. Tamaru, *J. Phys. Chem.*, 1955, **59**, 1084.
49. 'Vapor Deposition', p. 283.
50. T. L. Chu *et al., CVD*, 1975, **V**, 653.
51. CVD of Si, Semiconductor Silicon 1977, Electrochem. Soc., Princeton, **NJ**, 1977.
52. A. K. Praturi, *CVD*, 1977, **VI**, 20.
53. S. Andisio, *J. Electrochem. Soc.*, 1980, **127**, 2299.
54. C. F. Powell *et al., J. Electrochem. Soc.*, 1948, **93**, 258.
55. I. E. Campbell *et al., J. Electrochem. Soc.*, 1948, **93**, 271.
56. B. W. Gosner, Titanium Rep. Symp., ONR, Washington, DC, 1948.
57. G. Derge and G. P. Monet, US 2 743 173, 1956.
58. J. Spitz and J. Chevallier, *CVD*, 1975, **V**, 204.
59. C. M. Hollabaugh *et al., CVD*, 1977, **VI**, 559.
60. A. E. van Arkel, 'Reine Metalle', Springer, Berlin, 1939.
61. A. E. van Arkel *et al.*, US 1 891 124, 1932.
62. A. Bremer and W. E. Reid, US 3 072 983, 1963.
63. C. F. Powell *et al., J. Electrochem. Soc.*, 1948, **93**, 258.
64. J. J. Lander, US 2 516 058, 1930.
65. Z. M. Shapiro, 'Metallurgy of Zirconium', McGraw-Hill, NY, 1955.
66. P. J. Sherwood *et al., CVD*, 1975, **V**, 801.

67. A. M. Shroff *et al., CVD*, 1975, **V**, 351.
68. J. A. Papke and R. D. Stevenson, *CVD*, 1969, **I**, 193.
69. K. J. Miller *et al., J. Electrochem. Soc.*, 1966, **113**, 902.
70. A. R. Badzian and T. Badzian, *Surf. Coat. Technol.*, 1988, **36**, 283.
71. S. Furuno *et al.*, JP 6,381,810, April 1988.
72. V. Hoffe and A. Tehel, *Wiss. Tag. Tech. Univ. Karl-Marx-Stadt*, 1987, 201.
73. A. Sawabe *et al., Appl. Phys. Lett.*, 1987, **50**, 728.

Table 35.3 PHYSICAL VAPOUR DEPOSITION OF OXIDES

Compound	*Process**	*Coating thickness* (μm)	*Typical applications*	*Ref.*
Al_2O_3	S	1–6	Passive sealing layers	—
	RSP		Wear-resistant layers	1, 2
	IP			
CuO/Cu_2O	r.f. S Cu + O_2	—	—	3
Cr_2O_3	r.f. S	0.1	Wear resistant layer	4, 5
	RE	0.01	Anti-reflectance coating	
NiO	S	1	—	6
SiO_2	MS	0.006	Mask layer on IC	—
	r.f. S	13	Multilayer laser fusion test	7, 8
SiO–Cr	r.f. S	—	Resistor film	9
SnO_2	RSP	—	Transparent electrodes for electro-optics	10
Ta_2O_5	r.f. S Ta + O_2	—	Optical wave guides	11
TiO_2	RSP Ti + O_2	1–2	Anti-reflectance coatings on Al mirror	12
ZnO	r.f. S	—	Surface acoustic wave (SAW) devices	13
ZrO_2 (Y-stabilised)	d.c./r.f. S	25–100	Thermal barrier coatings for gas-turbine parts	14
	EB			15

* *See* Table 35.1 for the key

REFERENCES TO TABLE 35.3

1. R. S. Nowicki, *J. Vac. Sci. Tech.*, 1977, **14**, 127.
2. R. F. Bunshah and R. J. Schramm, *Thin Solid Films*, 1977, **40**, 211.
3. M. Samirant *et al., Thin Solid Films*, 1971, **8**, 293.
4. B. Bhushan, *Thin Solid Films*, 1979, **64**, 231.
5. E. Ritter *et al.*, US 4 172 156, 1976.
6. P. V. Plunkett *et al., Thin Solid Films*, 1979, **64**, 121.
7. K. Urbanek, *Solid State Tech.*, 1977, **20**, 87.
8. S. F. Meyer and E. J. Hsieh, *Thin Solid Films*, 1979, **64**, 383.
9. V. Fronz *et al., Thin Solid Films*, 1980, **65**, 33.
10. E. Leja *et al., Thin Solid Films*, 1980, **67**, 45.
11. W. M. Paulson *et al., J. Vac. Sci. Tech.*, 1979, **16**, 307.
12. L. D. Hartsough and P. S. McLeod, *J. Vac. Sci. Tech.*, 1977, **14**, 123.
13. K. Ohji *et al., J. Vac. Sci. Tech.*, 1978. **14**, 123.
14. J. W. Patten, *Thin Solid Films*, 1979, **64**, 337.
15. A. S. James *et al., Surf. Coat. Technol.*, 1987, **32**, 377.

Table 35.4 CHEMICAL VAPOUR DEPOSITION OF OXIDES

Material	*Process*	*Temp.* (°C)	*Coating thickness* (μm)	*Typical applications*	*Ref.*
Al_2O_3	$AlCl_3 + CO_2 + H_2$	850–1 800	2–8	Hard coating for tool tips Coating for nuclear fuels	1–5
	$AlCl_3 + O_2$ plasma assisted	230–350		Passive sealing layer for integrated circuits. Single crystal growth	
	$Al(CH_3)_3$ or $Al(OC_3H_7)_3 + O_2$	275–500			
B_2O_3–SiO_2	$B_2H_6 + SiH_4 + O_2$	350	0.4–0.6	Diffusion source for doping electronic materials. Cladding for optical fibres	23
Cr_2O_3	$Cr(CO)_6 + O_2$	400–600	<25	Oxidation-resistant coating	6
Fe_2O_3	$Fe(acac)_3 + O_2$ plasma assisted	400–600	—	—	24
SiO_2	$SiH_4 + O_2$ (+*hν*)	300–430	5–25	Passive layer-encapsulant for electronics	7–12
	$Si(OEt)_4$ thermal decomp.	800–1 000		Anti-reflectance coating	25
	$Si(OEt)_4 + H_2O$	500–1 000		Corrosion-resistant coating for chemical plant. AGR plant	
	$SiH_4 + N_2O$	300–1 500			
	$SiCl_4 + O_2$ plasma assisted	>1 000	20–50	Prep. of optical fibres	
SnO_2	$SnCl_4 + O_2$	600–1 000	0.1	Semiconductor assemblies. Optoelectronic applications.	13–15
	$SnCl_4 + H_2O$	>430		Transparent electrically conducting coatings	
	$Sn(CH_3)_4 + O_2$	400–600	0.1		
TiO_2	$TiCl_4 + O_2$ + hydrocarbon (in flame)	450–800	—	Pigment production	16–19
	$TiCl_4 + CO$ plasma assisted or $TiCl_4 + O_2$ plasma assisted			Dielectric films in MOS structure	
	$Ti(OC_3H_7)_4 + O_2$	450			26
$YBa_2Cu_3O_{7-x}$	Y/Ba/Cu(acac) + O_2	400–600	1–20	Superconductor with high T_c	27, 28
ZrO_2	$Zr(acac)_4 + O_2$	450–700	—	MOS devices	20–22
	$ZrCl_4 + CO_2 + CO + H_2$	1 000	5–10	Hard coatings on sintered carbides. Oxidation-resistant coatings	
	$Zr(OC_3H_7)_4$	500–600	12–75		

REFERENCES TO TABLE 35.4

1. R. Funk *et al., J. Electrochem. Soc.*, 1976, **123**, 285.
2. J. N. Lindström and R. T. Johannesson, *J. Electrochem. Soc.*, 1976, **123**, 555.
3. P. S. Schaffer, *J. Am. Ceram. Soc.*, 1965, **48**, 508.
4. H. Katto and Y. Koga, *J. Electrochem. Soc.*, 1971, **118**, 1619.
5. M. T. Duffy and W. Kern, *RCA Rev.*, 1970, **31**, 754.
6. J. J. Lander, US 2 671 739, 1954; 'Vapor Deposition', p. 390.
7. J. Graham, *High Temperatures—High Pressures*, 1974, **6**, 577.
8. J. Middelhoek and A. J. Klinkhamer, *CVD*, 1975, **V**, 19.
9. 'Vapor Deposition', p. 392.
10. J. Irven and A. Robinson, *Phys. Chem. of Glass*, 1980, **21**, 47.
11. D. Küppers and H. Lydtin, *CVD*, 1977, **VI**, 461.
12. J. Irven and A. Robinson, *Electronic Letters*, 1979, **15**, 252.
13. O. Tabata, *CVD*, 1975, **V**, 681.
14. 'Vapor Deposition', Wiley, NY, p. 398, 1966.

15. R. N. Ghostagore, *J. Electrochem. Soc.*, 1971, **118**, 1619.
16. A. Pechukas and G. Atkinson, US 2 394 633, 1946.
17. H. F. Sterling *et al., Vide*, 1966, **21**, 80.
18. J. H. Alexander *et al.*, 'Thin Film Dielectrics', F. Vratny, Electrochem. Soc., NY, 1969.
19. C. C. Wang *et al., RCA Rev.*, 1970, **31**, 728.
20. B. Balog *et al., J. Cryst. Growth*, 1972, **17**, 298.
21. J. N. Lindström *et al.*, US 3 837 896, 1974.
22. K. S. Kagdiyasin and C. T. Lynch, USAF Rept. ASD-DR-322, 1963.
23. J. Wong, *J. Electrochem. Soc.*, 1980, **127**, 62.
24. A. Fuji *et al.*, JP 6 394 312, May 1988.
25. K. Ito *et al.*, JP 6 314 872, January 1988.
26. Y. Hokari, JP 6 372 883, April 1988.
27. H. Yamane *et al., Chem. Lett.*, 1988, 939.
28. A. D. Berry *et al., Appl. Phys. Lett.*, 1988, **52**, 1743.

Table 35.5 PHYSICAL VAPOUR DEPOSITION OF NITRIDES

Compound	*Process**	*Coating thickness* (μm)	*Typical applications*	*Ref.*
AlN	RIS Al + N_2	—	Electronic material	1–3
	RIP Al + N_2 or NH_3—substrate 1 000°C	2	Refractory dielectric Acoustic transducers	
	RE Al + NH_3	0.6–1.5	SAW diodes	
CrN	RIP Cr + N	5	Die casting dies	13
HfN	RSP	—	Thin film dielectrics	4
	RIP Hf + N_2	0.2–0.3		
NbN	RSP	—	Superconducting layers	5
TaN	RSP	115 Å min^{-1}	Resistor film	6
Si_3N_4	RIS Si + N_2	0.25–0.08	Encapsulant for GaAs	1,
	r.f. RSP Si + N_2		Optical wave guides	7–9
	RSP Si_3N_4 + N_2			
TiN	RSP Ti + N_2	—	Hard coatings on steel tools	10–11
	ARE Ti + N_2	5–15		
	RAE Ti + N_2			
W_2N	RSP	100 Å min^{-1}	Superconductivity	6
ZrN	RIP Zr + N_2	3–5	Wear resistant coatings	12

* *See* Table 35.1 for key.

REFERENCES TO TABLE 35.5

1. H. J. Erler *et al., Thin Solid Films*, 1980, **65**, 233.
2. Y. Murayama *et al., J. Vac. Sci. Tech.*, 1980, **17**, 796.
3. S. Yoshida *et al., Appl. Phys. Lett.*, 1975, **26**, 461.
4. F. T. J. Smith, *J. Appl. Phys.*, 1970, **41**, 4227.
5. J. Spitz and A. Aubert, *CVD*, 1975, **V**, 258.
6. F. M. Kilbane and P. S. Habig, *J. Vac. Sci. Tech.*, 1975, **12**, 107.
7. L. E. Bradley and J. S. Sites, *J. Vac. Sci. Tech.*, 1979, **16**, 189.
8. F. H. Eisen *et al., Solid State Electronics*, 1977, **20**, 219.
9. W. M. Paulson *et al., J. Vac. Sci. Tech.*, 1979, **16**, 307.
10. T. Abe and T. Yamashiua, *Thin Solid Films*, 1975, **30**, 19.
11. K. Nakamura *et al., Thin Solid Films*, 1977, **40**, 155.
12. O. Knofek *et al., Surf. and Coat. Techn.*, 1994, **68/69**, 489.
13. O. Knofek *et al., Surf. and Coat. Techn.*, 1993, **62**, 630.

Table 35.6 CHEMICAL VAPOUR DEPOSITION OF NITRIDES

Material	*Process*	*Temp.* (°C)	*Coating thickness* (μm)	*Typical applications*	*Ref.*
AlN	$AlCl_3 + NH_3 + H_2$	800–1 000	0.5	Electronic applications Refractory dielectrics SAW devices	1–5
	$Al + N_2$	1 800–2 000	Whiskers	High strength refractories	
	$AlCl_3 + N_2 + H_2$ plasma assisted	800–1 200		III/V semiconductors	
	$(CH_3)_3Al + NH_3 + H_2$	1 200	Epitaxial		
BN	$BCl_3 + NH_3$	1 000–2 000	1–2 000	Formation of free-standing crucibles, tubes and plates	6–12
	$BF_3 + NH_3$	1 000–2 000		Travelling wave tube isolators Insulating layers in semi-cond. manufacture	
	$B_2H_6 + NH_3$ plasma assisted	400–700	0.2–0.6	Diffusion doping of Si	
	$B_2H_6 + N_2$ microwave plasma assisted	800	4	Cubic BN for wear resistance	24
HfN	$HfCl_n + N_2 + H_2$ ($n = 2, 3, 4$)	900–1 300	5–10	Wear-resistant coatings Diffusion barriers	13
TaN	$TaCl_5/TaBr_5 + N_2$	800–1 500	—	Resistor film	14–15
Si_3N_4	$SiH_4 + N_2/NH_3$ plasma assisted	350	0.15	Encapsulation of III/V semiconductor devices	
	$SiF_4 + NH_3$	1 000–1 500		Anti-reflection coatings on solar cells	12, 16–20
	$SiCl_4 + NH_3$	1 000–1 500	10–2 000		
	$SiHCl_3 + NH_3$	1 100–1 400		Gas turbine parts	
	$SiH_4 + NH_3 + N_2H_4$ chem. decomp.				
TiN	$TiCl_4 + N_2 + H_2$	650–1 700	2–8	Hard coatings for hard-metal cutting tools. Scratch-resistant decorative surfaces. Diffusion barrier layers	21–23, 25
	$Ti(NMe_2)_4$ thermal decomp.	300–500			
	$TiCl_4 + N_2 + H_2$ plasma assisted	<550			
ZrN	$ZrCl_4 + N_2 + H_2$	900–1 200	2–8	Hard coatings for cutting tools	13

REFERENCES TO TABLE 35.6

1. A. J. Noreika and D. W. Ing, *J. Appl. Phys.*, 1968, **39**, 5578.
2. T. L. Chu and R. W. Kelm, Jr., *J. Electrochem. Soc.*, 1975, **122**, 995.
3. K. M. Taylor and C. Lenie, *J. Electrochem. Soc.*, 1960, **107**, 308.
4. J. Bauer *et al., Physica Status Solid* (a), 1977, 39, 173.
5. H. M. Manasevit *et al., J. Electrochem. Soc.*, 1971, **118**, 1864.
6. G. Clerc and P. Gerlach, *CVD*, 1975, **V**, 777.
7. D. Morin and M. Le Clercq, French Pat. 2 232 613, 1975.
8. N. J. Archer, High Temp. Chem. of Inorg. and Ceram. Mats., Chem. Soc. Pub. 30, 1977.
9. H. O. Pierson, *J. Compos. Mat.*, 1975, **9**, 228.
10. R. Francis and E. P. Flint, US Army Report, WAL-766, 41/1, 1961.
11. M. Hirayama and K. Shohno, *J. Electrochem. Soc.*, 1975, **122**, 1671.
12. K. Shohno et al., *J. Electrochem. Soc.*, 1980, **127**, 1546.
13. M. J. Hakim, *CVD*, 1975, **V**, 634.
14. R. Kieffer *et al., Powder Metall. Inst.*, 1973, **5**, 188.
15. K. Hieber, *Thin Solid Films*, 1974, **5**, 188.
16. Dietrich and Reid, *Ext. Abst., Electrochem. Soc.*, 1977, **77-2**, 510.
17. M. Ružička, *Czech. J. Phys.*, 1974, **B24**, 465.
18. J. Gebhart *et al., CVD*, 1975, **V**, **786**.
19. J. Bühler *et al., CVD*, 1977, **VI**, 493.
20. Hughes Aircraft US Pub. 4 181 751, 1980.
21. W. Schintlmeister *et al., CVD*, 1975, **V**, 523.
22. K. Sugiyama and S. Motojima, *CVD*, 1975, **V**, 147.
23. R. Warren and M. Carlsson, *CVD*, 1975, **V**, 611.
24. K. Fukushima and M. Tobioka, JP 6 340 800, February 1988.
25. T. Arai *et al., Thin Solid Films*, 1988, **165**, 139.

Table 35.7 PHYSICAL VAPOUR DEPOSITION OF CARBIDES

Compound	*Process**	*Coating thickness* (μm)	*Typical applications*	*Ref.*
Cr_3C_2	RSP IP	0.3–2.5	Solar energy collectors Wear-resistant coatings	1–2
Cr_7C_3	ARE			
HfC	ARE/RE $Ar + C_2H_2 + Hf$	2.5 μm min^{-1}	—	2
Mo_2C	RSP IP $Ar + CH_4$	0.18	Solar energy absorption	1
NbC	ARE/RE $Ar + CH_4$ or $C_2H_4 + Nb$	2.5 μm min^{-1}	Optical/semiconductor	2
SiC	RSP $Ar + C_2H_2$	1	Electronic barrier coating	3–4
Ta_2C	RSP	0.8	Solar energy absorption	1, 2
TaC	$Ar + CH_4$, C_2H_2 ARE/RE $Ta + C_2H_2$	— 1.5 μm min^{-1} at 590°C	—	— 5
TiC	RSP A + CH4 ARE ARE/RE $Ti + C_2H_4$	5.0 μm at 700/ 900°C 4.0 μm min^{-1} at 450°C	Cutting tools Wear-resistant surfaces	2 6,7,8
VC	ARE $V + C_2H_2$	3.0 μm min^{-1} at 555°C	—	2
W_2C	RSP SIP W/C	0.8	Solar energy absorption Drill tips, pump components	1 10
WC	$W + Ar + CH_4$	—	—	—
ZrC	ARE/RE $Zr + C_2H_2$	5.0 μm min^{-1} at 540°C	Diffusion barrier	2, 9

* *See* Table 35.1 for key.

REFERENCES TO TABLE 35.7

1. G. L. Harding, *J. Vac. Sci. Tech.*, 1976, **13**, 1070.
2. R. F. Bunshah and A. C. Raghuram, *J. Vac. Sci. Tech.*, 1972, **9**, 1385.
3. K. E. Haq, *Appl. Phys. Lett.*, 1975, **26**, 255.
4. Y. Murayama and T. Takao, *Thin Solid Films*, 1977, **40**, 309.
5. W. Grossklaus and R. F. Bunshah, *J. Vac. Sci. Tech.*, 1975, **12**, 811.
6. F. Shinoki and A. Itoh, *Jap. J. Appl. Phys. Suppl.*, 1974, **2** (1), 505.
7. K. Nakamura *et al., Thin Solid Films*, 1977, **40**, 155.
8. W. R. Stowell, *Thin Solid Films*, 1974, **22**, 111.
9. K. D. Kennedy and Scheuermann, US 3 900 592, 1975.
10. J. P. Coad *et al.*, BP 2 174 678, March 1987.

Table 35.8 CHEMICAL VAPOUR DEPOSITION OF CARBIDES

Material	*Process*	*Temp.* (°C)	*Coating thickness* (μm)	*Typical applications*	*Ref.*
B_4C	$BCl_3 + CO + H_2$ $B_{10}C_2H_{12}$ thermal decomp. $B_2H_6 + CH_4$ BMe_3 thermal decomp. $BCl_3 + CCl_4$ thermal decomp. $BCl_3 + CH_4 + H_2$	1 200–1 800 1 000–1 300 ~550 1 200–2 000 1 300–1 900	 10–1 000	Nuclear industry Armament Abrasive blasting nozzles Rocket nozzles Fibres	1–6
Cr_3C_2 Cr_7C_3	$Cr(CO)_6 + H_2$ $CrCl_2 + CH_4 + H_2$ $CrCl_2 + H_2$ Cr dicumene–thermal decomp.	300–650 ~1,000 1 000 450–650	 10–40 	Wear-resistant coatings HV 2250 Chromising of steel	7 8 9 10
HfC	$HfCl_4 + H2 + CH_4$	1 000–1 500	5	Diffusion barrier Wear resistant coating	11 12
Mo_2C	$Mo(CO)_6$ thermal decomp. $MoF_6 + C_6H_6 + H_2$	350–475 400–1 000	—	Wear-resistant layer	13 14

(*continued*)

Table 35.8 CHEMICAL VAPOUR DEPOSITION OF CARBIDES—*continued*

Material	*Process*	*Temp.* (°C)	*Coating thickness* (μm)	*Typical applications*	*Ref.*
NbC	$NbCl_5 + CCl_4 + H_2$	1 500–1 900	—	Wear-resistant coating Diffusion barrier	15
SiC	$SiCl_4 + CH_4$	1 000–2 000	10–1 000	Oxidation-resistant coatings Single fibres of high strength	16
	$CH_3SiCl_3 + H_2$	1 000–1 600		Heating elements Electrical heat stylus	17
	$SiH_4 + C_2H_4$ or CH_4 plasma assisted		~1	Electrochemical machining tool	18
Ta_2C TaC	$TaCl_5 + CH_4$	1 100		Coating W filaments for incorporation into superalloy matrices	19
TiC	$TiCl_4 + H_2 + CH_4$ or C_6H_6 $TiCl_4 + CCl_4 + H_2$ Ti organic	800–1 400 >1 000 150–500	2–12	Wear-resistant coatings on steel and carbides— particularly cutting tools	20–24 35 36
VC	$VCl_2 + CH_4 + H_2$ Diffusion coating VC_x ($x = 0.84$–0.89)	1 050–1 130	—	Wear-resistant coating	25–26
WC W_2C	$WF_6 + C_6H_6 + H_2$ $W(CO)_6$ thermal decomp. $WF_6 + CO + H_2$ $WCl_6 + CH_4 + H_2$	400–900 300–500 600–1 000 900–1 150	— 5–50	Wear-resistant coatings	27–31
ZrC	$ZrCl_4 + CH_4 + H_2$	1 050–1 500	50–100	Wear-resistant coating Diffusion barrier layer	32–33
	$ZrCl_4 + C_3H_6 + H_2$	1 150–1 300	50	High temperature insulation Nuclear fuel coating	34

REFERENCES TO TABLE 35.8

1. M. Formstecher and E. Ryskevic, *Compt. rend.*, 1945, **221**, 558.
2. R. L. Hough, *SAMPE*, 1979, **10** (1), 25.
3. G. R. Martin, US 2 484 519, 1949.
4. As 3.
5. S. Mierzejewska and T. Niemyski, *J. Less-Common Met.*, 1965, **8**, 368.
6. R. G. Bourdean, US 3 334 967, 1967.
7. 'Vapor Deposition', p. 360.
8. B. B. Owen and R. T. Weber, Am. Inst. Min. Met. Tech. Pub. 2306, 1948.
9. S. Csch, Proc. 3rd Int. Symp. Met. and Heat Treatment Met., Warsaw, 1967.
10. J. E. Gates *et al.*, US 3 951 612, 1980.
11. M. J. Hakim, *CVD*, 1975, **V**, 634.
12. D. Hertz *et al.*, *High Temp.–High Pressure*, 1974, **6**, 423.
13. 'Vapor Deposition', p. 362.
14. R. H. Lewin and C. Hayman, BP 1 326 769.
15. T. A. Lyndvinskaya *et al.*, 'Refractory Carbides', N. Y. Consultants Bur., 1974.
16. E. Fitzer *et al.*, *CVD*, 1975, **V**, 523.
17. H. Beutler *et al.*, *CVD*, 1975, **V**, 749.
18. G. Verspui, *CVD*, 1977, **VI**, 366.
19. W. J. Heffernan *et al.*, *CVD*, 1973, **IV**, 498.
20. W. Schintlemeister *et al.*, *CVD*, 1979, **V**, 523.
21. M. Maillat *et al.*, *Thin Solid Films*, 1979, **64**, 243.
22. Many papers, *CVD*, 1972, **III**.
23. V. K. Sarin and J. N. Lindstrom, *CVD*, 1977, **VI**, 389.
24. J. J. Nickl *et al.*, *J. Less-Common Met.*, 1972, **26**, 335.
25. G. Ebersbach *et al.*, *Die Technik*, 1974, **29**, 273.
26. E. Horvath and A. J. Perry, *Thin Solid Films*, 1980, **65**, 309.
27. N. J. Archer, *CVD*, 1975, **V**, 556.

28. N. J. Archer and K. K. Yee, *Wear*, 1978, **48**, 237.
29. 'Vapor Deposition', p. 372.
30. D. A. Tarver, US 3 574 672, 1971.
31. H. Mantle *et al.*, *CVD*, 1975,**V**, 540.
32. T. C. Wallace, *CVD*, 1973, **IV**, 91.
33. C. Hollabaugh *et al.*, *CVD*, 1977, **VI**, 419.
34. A. R. Driesner *et al.*, *CVD*, 1973, **IV**, 473.
35. A. Inzenhofer, *Jud.-Anz.*, 1988, **11**, 28.
36. A. E. Kaloyeros *et al.*, *Rev. Sci. Instrum.*, 1988, **59**, 1209.

35.3 Thermal spraying

Thermal spraying is a process in which a material is heated to a molten, partly-molten or highly plastic solid state by a heat source and is accelerated to the surface on a substrate by a gas jet. On impacting the substrate, the spray material droplets flatten, solidify and cool down on the surface or overlap on the fore-flattened droplets. In this way, the spray material is deposited to form a coating. Figure 35.1 shows the principle of thermal spraying schematically. A wide range of spray materials from ceramics to metal alloys can be deposited to protect a component against wear, corrosion, oxidation or heat or to generate special electric, magnetic or biomedical properties. Table 35.9 gives some of spray materials used commonly. The significant advantages of thermal spraying are:

1. Coatings of nearly all materials can be deposited on various substrates
2. Low heat load on the substrate material
3. Local coating is possible
4. Some facilities are mobile
5. High deposition rate.

Thermal spraying processes are usually distinguished by the heat sources used. The most widely used techniques are conventional combustion spraying, high velocity oxy-fuel spraying, wire arc spraying and plasma spraying. There are different processes variants under each one of the above mentioned heat sources.[1]

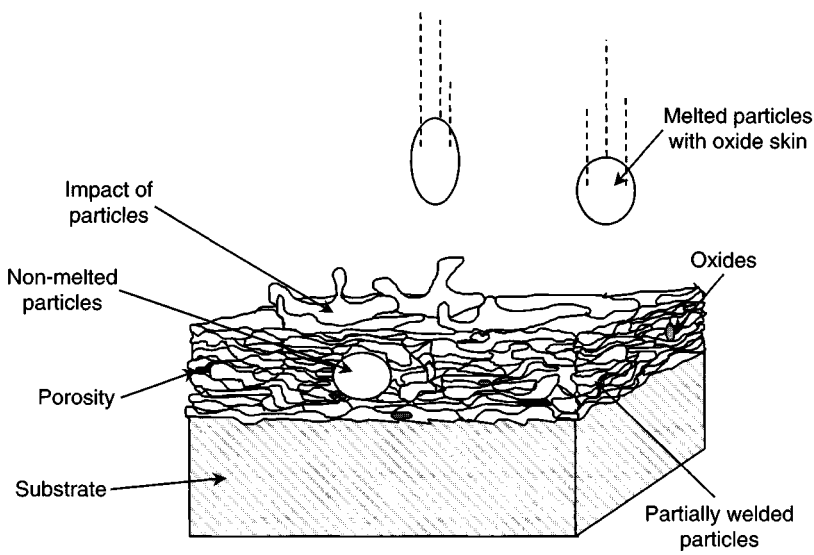


Figure 35.1 *Principle of deposition of a thermal sprayed coating*

Table 35.9 EXAMPLES FOR COATING MATERIALS FOR THERMAL SPRAYING[1,4]

Material group	*Example*
Pure metals	Al, Co, Mo, Ni, Ti, Zn,...
Alloys	Ni–Al, Ni–Cr, Ni–Cr–Al, Steel, ...
Pseudo-Alloys	Cu–W, Bronze-Steel
Oxide-Ceramics	Al_2O_3, Cr_2O_3, TiO_2, ...
Hard materials	Cr_3C_2/Ni, WC/Co, ...
Synthetics	EMAA, ETFE,...

For most thermal spray processes, the particles are heated up very fast and can react with the gases of the surrounding atmosphere. Because of this, the material of the particles can change its characteristics during the spraying process. The different effects are selective evaporation of components, reaction of metal compounds and formation of non-volatile metal compounds like oxides, nitrides, and hydrides.[5]

35.3.1 Combustion wire spraying

In the combustion wire spray process the spray material in wire form is fed continually into a flame where it is melted. The spike of the wire is atomised by the surrounding compressed air. The flame reaches temperatures up to 3 160°C. The accelerated particles deposit on the workpiece surface. Typical fuel gases are acetylene, propane and hydrogen.[1]

35.3.2 Combustion powder spraying

The powder flame spray process is similar to the combustion wire spray process with the difference, that powder and not a wire is injected into the flame. This offers a much wider range of coating material options than the wire flame spray process. The powdered material is fed continually into the fuel gas-oxygen flame where it is typically melted or semi-melted by the heat of combustion. A powder feed carrier gas transports the powder particles into the combustion flame, and the gases transport the material towards the prepared workpiece surface. Typical choices for fuel gases are acetylene or hydrogen.[1]

35.3.3 Electric wire arc spraying

Two identical or different metallic wires are moved into a spray gun with a controlled speed. The two wires are electrically charged with opposing polarity. When the wires are brought together at the nozzle the arising arc melts the material. Compressed air atomises and accelerates the molten material onto the workpiece surface. The temperatures in the electric arc are approximately 6 500°C. This process can also be driven with three or four wires in the same way.[1]

35.3.4 High velocity oxy-fuel spraying (HVOF)

High velocity oxygen fuel spraying is a further development of the combustion powder spray process. HVOF has reached a good market position in a relatively short time. HVOF is characterised by very high gas and particle velocities. A reaction of oxygen and fuel gas in the combustion chamber generates the heat required for melting the powder. The expanding gas accelerates the particles to velocities up to 300–600 m/s.

Low particle temperature and high particle speed are the advantages of the HVOF process for a lot of materials and applications. The coatings are characterised by a low porosity, high adhesion, high hardness and a good resistance against wear.[1]

35.3.5 Plasma spraying

Plasma spraying is a very flexible thermal spray process. It can provide energy in the form of a plasma jet with temperatures higher than 20 000°C to melt any material. The number of coating materials

that can be deposited with the plasma spray process is almost unlimited. For the plasma spray process the plasma is generated by a arc between two water cooled electrodes in a torch (Figure 35.2). The atoms of the plasma gas are ionised when passing the arc. For a diatomic gas molecular dissociation will also happen. As plasma gas argon, helium, hydrogen, nitrogen and any mixture of these gases are used. During the ion/electron recombination process heat is released. The coating material is injected into the gas plume, melted, accelerated and deposited on a surface.

The sensitivity of the plasma spraying process to operating conditions and the great number of adjustable parameters makes the optimisation of the process for each application necessary.[2]

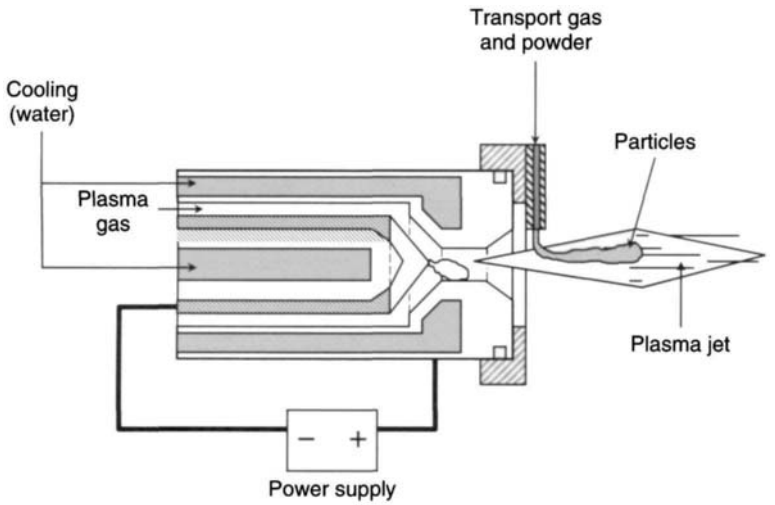


Figure 35.2 *Plasma spraying process*

Table 35.10 THERMAL SPRAY PROCESSES, CHARACTERISATION AND TYPICAL APPLICATIONS

Process	*Process temperature/ particle velocity/ deposition rate*	*Typical applications*
Combustion powder spray	<3 160°C/<50 m/s/ 1–6 kg/h	Repair operations Resistance to impact Abrasion and wear resistance Resistance to chemical attack Resistance to atmospheric and heat corrosion
Combustion wire spray	<3 160°C/<200 m/s/ 6–8 kg/h	Resistance to atmospheric and heat corrosion Repair operations
Electric wire arc spray	≈4 000°C/≈150 m/s/ 8–20 kg/h	Resistance to atmospheric and heat corrosion Repair operations Bond coats Electrically conductive and solderable coatings Anti-skid and traction coatings
High velocity oxy-fuel thermal spray (HVOF)	<3 160°C/<550 m/s/ 2–8 kg/h	Repair operations Abrasion, wear or erosion resistance Sliding wear resistance Resistance against cavitation Resistance to chemical attack Control of oxidation and sulphidation Resistance to atmospheric and heat corrosion

(*continued*)

Table 35.10 THERMAL SPRAY PROCESSES, CHARACTERISATION AND TYPICAL APPLICATIONS—*continued*

Process	*Process temperature/ particle velocity/ deposition rate*	*Typical applications*
Plasma spray	>20 000°C/<450 m/s/ 4–8 kg/h	Repair operations Abrasion or erosion resistance
		Sliding wear resistance Resistance to fretting, galling or adhesive wear Resistance to impact and cavitation Resistance to chemical attack Control of oxidation and sulphidation Resistance to atmospheric and heat corrosion Galvanic corrosion control Thermal or electrical insulation Abradables
Laser spraying	>10 000°C/>1 m/s	Partial coating of tolls
Detonation spraying	<3 160°C/ca. 600 m/s/ 2–8 kg/h	Abrasion or erosion resistance Sliding wear resistance Resistance to fretting, galling or adhesive wear Resistance to cavitation Resistance to chemical attack Control of oxidation and sulphidation

Table 35.11 FUNCTIONS AND PROPERTIES OF SELECTED THERMAL SPRAYED COATINGS[3]

Coating	Resistance against...				Slide capability	Hardness [HV]	Maximum application temperature [°C]
	Corrosion	Oxidation	Wear abrasive	Adhesive			
Aluminium	++[1]						
Zinc	++[1]						
Molybdenum	+	+	++	++	++	1 000	350
Steel	+					250–500	[2]
MCrAlY	+++	+++					1 000
Hard alloy	+		++			600	900
Tribaloy	++		++	++	++	600	1 000
WC/Co			++	+++	++	1 000	540
CrC/NiCr			+++	+++	++	800	1 000
Alumina			++	+++	+	2 000	[2]
Chromium oxide			++	+++	++	2 300	[2]
Zirconia (stabilised)			++	+++		1 500	
Co-alloy/ Chromium oxide			+++	+++		600	1 000
Nickel/graphite					[4]	[3]	500

[1] cathodic protection, [2] depends on alloy, [3] depends on porosity of the coating on steel, [4] abradable.
+: low, ++: medium, +++: high.

REFERENCES TO SECTION 35.3

1. Fr. W. Bach, T. Duda, Moderne Beschichtungsverfahren, 2000, WILEY-VCH Verlag, ISBN 3-527-30117-8.
2. E. Lugscheider, I. Kvernes, Thermal Barrier Coatings, Powder Spray Process and Coating Technology, Intermetallic And Ceramic Coatings, 1999, Marcel Dekker, Inc., ISBN 0-8247-9913-5.
3. H. Simon, M. Thoma, Angewandte Oberflaechentechnik für metallische Werkstoffe, Carl Hanser Verlag, 1985.
4. M. Allen, C. C. Berndt, J. A. Brogan, D. Ottersond, Thermal Sprayed Polymer Coatings for Corrosion Protection In A Biochemical Treatment Process, Proc. Of the 15th ITSC, 25–29 May 1998, Nice, France.
5. R. A. Haefer, Oberflaechen- und Duennschichttechnologie, Springer Verlag Berlin, 1987.